

K Grothendieck's Identity Beyond Gaussian Embeddings

In this section we summarize several facts about random sign embeddings used to approximate angular inner products. Grothendieck's identity is first extended from Gaussian vectors to arbitrary rotationally invariant distributions. This immediately permits Haar-orthogonal embeddings obtained from the polar factor of a Gaussian matrix; within orthogonal blocks, the estimator remains unbiased and has strictly smaller variance than its i.i.d. Gaussian counterpart. We then recall the approximate Rademacher identity of Aghasi et al. and use it to establish a compact expectation theorem for a full-width, one-pass randomized Walsh–Hadamard embedding.

K.1 Gaussian Grothendieck's Identity

Let $u, v \in \mathbb{R}^p$ be nonzero and let $g \sim \mathcal{N}(0, I_p)$. The random-hyperplane form of Grothendieck's identity is

$$\mathbb{E}[\text{sign}(g^\top u) \text{sign}(g^\top v)] = \frac{2}{\pi} \arcsin\left(\frac{u^\top v}{\|u\|_2 \|v\|_2}\right). \quad (26)$$

Consequently, if $W \in \mathbb{R}^{n \times p}$ has unit-norm rows, $x \in \mathbb{R}^p$ is a unit vector, and $\mathcal{G} \in \mathbb{R}^{N \times p}$ has i.i.d. standard Gaussian entries, then, entrywise,

$$\mathbb{E}\left[\frac{1}{N} \text{sign}(W\mathcal{G}^\top) \text{sign}(\mathcal{G}x)\right] = \frac{2}{\pi} \arcsin(Wx). \quad (27)$$

K.2 Rotational Invariance and Orthogonal Embeddings

Gaussianity is not essential in Eq. (26); only the random direction matters.

Theorem K.1 (Rotationally invariant Grothendieck identity). *Let $a \in \mathbb{R}^p$ be rotationally invariant, meaning $Ua \stackrel{d}{=} a$ for every orthogonal matrix U , and assume $\mathbb{P}(a = 0) = 0$. Then, for all nonzero $u, v \in \mathbb{R}^p$,*

$$\mathbb{E}[\text{sign}(a^\top u) \text{sign}(a^\top v)] = \frac{2}{\pi} \arcsin\left(\frac{u^\top v}{\|u\|_2 \|v\|_2}\right). \quad (28)$$

Proof. Write $a = R\theta$, where $R = \|a\|_2 > 0$ and θ is uniform on the unit sphere. The positive radius does not affect either sign. If α is the angle between u and v , rotational invariance reduces the calculation to their two-dimensional span. The two signs disagree on wedges of total angular proportion α/π . Hence the expectation is $1 - 2\alpha/\pi = (2/\pi) \arcsin(\cos \alpha)$. $\square$

This result allows embedding designs with better concentration than the general Gaussian matrices. Basically, rows of a random matrix need only be *marginally* rotationally invariant; independence is unnecessary for the mean. For instance, orthonormal random embeddings can offer a better concentration around the mean. In particular, let $\mathcal{G} \in \mathbb{R}^{N \times p}$ be standard Gaussian, with $N \geq p$, and define its polar factor

$$\mathcal{Q} = \mathcal{G}(\mathcal{G}^\top \mathcal{G})^{-1/2} \in \mathbb{R}^{N \times p}. \quad (29)$$

Then $\mathcal{Q}^\top \mathcal{Q} = I_p$ and $\mathcal{Q}$ is Haar distributed on the Stiefel manifold. Each row is marginally rotationally invariant, and therefore

$$\mathbb{E}\left[\frac{1}{N} \text{sign}(W\mathcal{Q}^\top) \text{sign}(\mathcal{Q}x)\right] = \frac{2}{\pi} \arcsin(Wx). \quad (30)$$

A strict finite-sample variance comparison follows directly for orthogonal blocks. Suppose $N = \nu p$ and form an $N \times p$ embedding by stacking ν independent square polar factors, each obtained from an independent $p \times p$ Gaussian matrix through Eq. (29). For fixed unit vectors u, v , let $\rho = u^\top v$. The variance of the estimator based on N independent Gaussian hyperplanes is

$$\text{Var}\left[\frac{1}{N} \sum_{j=1}^N \text{sign}(g_j^\top u) \text{sign}(g_j^\top v)\right] = \frac{1}{N} \left[1 - \left(\frac{2}{\pi} \arcsin \rho\right)^2\right]. \quad (31)$$

865 Within one orthogonal block, if write $Z_j = \text{sign}(q_j^\top u) \text{sign}(q_j^\top v)$, then exchangeability gives

$$\text{Var} \left[\frac{1}{p} \sum_{j=1}^p Z_j \right] = \frac{1}{p} \left[1 - \left(\frac{2}{\pi} \arcsin \rho \right)^2 \right] + \frac{p-1}{p} \text{Cov}(Z_1, Z_2). \quad (32)$$

866 The orthogonal-hyperplane variance theorem of Ji et al. [8]¹ shows that $\text{Cov}(Z_1, Z_2) < 0$ whenever
867 $-1 < \rho < 1$. Averaging ν independent blocks therefore yields

$$\text{Var} \left[\frac{1}{N} \text{sign}(W_i \mathcal{Q}^\top) \text{sign}(\mathcal{Q}x) \right] < \frac{1}{N} \left[1 - \left(\frac{2}{\pi} \arcsin(W_i x) \right)^2 \right], \quad (33)$$

868 which is exactly the variance of the corresponding estimator formed from an i.i.d. Gaussian matrix
869 in $\mathbb{R}^{N \times p}$. Here $\mathcal{Q}$ denotes the stacked orthogonal-block embedding; the single tall polar factor in
870 Eq. (29) retains the exact expectation in Eq. (30).

871 K.3 Rademacher Embeddings

872 The close behavior of Gaussian and Rademacher projections is quantified by the approximate
873 Grothendieck identity proved in the G-Net paper of Aghasi et al. [1]. Let $\tau \in \{\pm 1\}^p$ have independent
874 Rademacher entries. If u, v are unit vectors satisfying

$$\|u\|_\infty, \|v\|_\infty \leq \frac{C_0}{\sqrt{p}}, \quad |u^\top v| \leq 1 - \delta, \quad (34)$$

875 for fixed $C_0, \delta > 0$, then

$$\mathbb{E}[\text{sign}(\tau^\top u) \text{sign}(\tau^\top v)] = \frac{2}{\pi} \arcsin(u^\top v) + \mathcal{O}_{C_0, \delta}(p^{-1/2}). \quad (35)$$

876 Thus, if $\mathcal{R} \in \{\pm 1\}^{N \times p}$ is an i.i.d. Rademacher matrix and the rows of $W \in \mathbb{R}^{n \times p}$ and x satisfy
877 Eq. (34), then

$$\mathbb{E} \left[\frac{1}{N} \text{sign}(W \mathcal{R}^\top) \text{sign}(\mathcal{R}x) \right] = \frac{2}{\pi} \arcsin(Wx) + \mathcal{O}_{C_0, \delta}(p^{-1/2}) \quad (36)$$

878 entrywise. This result explains why dense Rademacher embeddings can closely track dense Gaussian
879 embeddings when the input and weight vectors are sufficiently dispersed.

880 K.4 Full-Width Randomized Walsh–Hadamard Embeddings

881 The preceding Rademacher approximation motivates structured matrices that retain sign randomness
882 while admitting fast multiplication. Let p be a power of two and let H_p be the normalized symmetric
883 Walsh–Hadamard matrix. Suppose $N = \nu p$. For $b = 1, \dots, \nu$, let

$$\mathcal{D}_b = \text{diag}(\mathcal{d}_{b1}, \dots, \mathcal{d}_{bp}), \quad \mathcal{d}_{bk} \stackrel{\text{i.i.d.}}{\sim} \text{Rad}, \quad (37)$$

884 and define the structured random matrix

$$\mathcal{A} = \begin{bmatrix} H_p \mathcal{D}_1 \\ \vdots \\ H_p \mathcal{D}_\nu \end{bmatrix} \in \mathbb{R}^{N \times p}. \quad (38)$$

885 Each block implements the one-pass transform $z^\top \mapsto z^\top \mathcal{D}_b H_p$.

Theorem K.2 (Full-width one-pass Hadamard expectation). *Let $W \in \mathbb{R}^{n \times p}$ have unit rows and let $x \in \mathbb{S}^{p-1}$ be unit norm. If the dispersion and nondegeneracy conditions in Eq. (34) hold for x and every row of W , then*

$$\mathbb{E} \left[\frac{1}{N} \text{sign}(W \mathcal{A}^\top) \text{sign}(\mathcal{A}x) \right] = \frac{2}{\pi} \arcsin(Wx) + \mathcal{O}_{C_0, \delta}(p^{-1/2}) \quad (39)$$

entrywise. The case $N = p$ is the single full-width, one-pass formula.

¹Ji, Jianqiu, et al. "Super-bit locality-sensitive hashing." Advances in neural information processing systems 25 (2012).

887 *Proof.* Let h_j be the j -th column of H_p . For every fixed b and j , $\sqrt{p}\mathcal{D}_b h_j$ is an i.i.d. Rademacher
 888 vector. Hence, for each row W_i ,

$$\begin{aligned} & \mathbb{E} \left[\frac{1}{N} \sum_{b=1}^{\nu} \sum_{j=1}^p \text{sign}(W_i \mathcal{D}_b h_j) \text{sign}(h_j^\top \mathcal{D}_b x) \right] \\ &= \mathbb{E}_i [\text{sign}(z^\top W_i^\top) \text{sign}(z^\top x)]. \end{aligned}$$

889 Applying Eq. (35) proves the claim. Independence among the coordinates within a Hadamard block
 890 is not needed for the expectation. $\square$

891 Each block uses one random sign diagonal and one fast Walsh–Hadamard transform, requiring
 892 $O(p \log p)$ arithmetic per vector. The construction is therefore a structured $N \times p$ analogue of the
 893 dense Rademacher embedding with the same population expectation up to the $O(p^{-1/2})$ invariance
 894 error.

895 **Implementation alternatives.** The same full-width formula can be used after zero-padding a vector
 896 from d coordinates to the next power-of-two width L :

$$z^\top \mapsto \text{sign}(\text{Pad}_L(z)^\top \mathcal{D} H_L).$$

897 Other efficient variants include a block-diagonal Hadamard transform and the one- or two-pass maps

$$\mathcal{T}_{\text{one}} = \mathcal{D}_1 H_{\text{blk}}, \quad \mathcal{T}_{\text{two}} = \frac{1}{\sqrt{b}} \mathcal{D}_1 H_{\text{blk}} \mathcal{D}_2 H_{\text{blk}},$$

898 where $H_{\text{blk}} = I \otimes H_b$. These are useful implementation choices when the hardware favors fixed-
 899 size Walsh–Hadamard blocks. Theorem 2 applies directly to the full-width one-pass construction;
 900 blockwise and two-pass forms can be treated as structured alternatives whose expectation depends on
 901 their additional block and mixing structure.

902 To summarize, Grothendieck’s identity is a consequence of uniform random direction and hence holds
 903 for every nondegenerate rotationally invariant distribution. Gaussian polar factors provide orthogonal
 904 random directions with the same exact mean and strictly smaller variance within orthogonal blocks.
 905 For an easier construction of embedding matrices, Rademacher vectors recover the same expectation
 906 up to $O(p^{-1/2})$ for dispersed inputs, and a full-width randomized Walsh–Hadamard transform
 907 inherits a similar expectation. This sequence offers several alternatives, ranging from exact Gaussian
 908 constructions to computationally efficient structured sign embeddings.